Machine Learning Presentation

By Kaiden Fury

Code for reading the dataset and missing values

File Edit Selection View Go Run ··· numpy test

Fury_Final_Project.ipynb

final Project > Fury_Final_Project.ipynb > M↓ KNN

Generate + Code + Markdown | Run All | Restart | Clear All Outputs | Jupyter Variables | Outline ··· numpy test (3.11.14) (Python 3.11.14)

[2] ✓ 0.0s Python

```
df.head()
```

[3] ✓ 0.0s Python

	Age	Gender	Income	Education	Marital Status	Number of Children	Home Ownership	Credit Score
0	25	Female	50000	Bachelor's Degree	Single	0	Rented	High
1	30	Male	100000	Master's Degree	Married	2	Owned	High
2	35	Female	75000	Doctorate	Married	1	Owned	High
3	40	Male	125000	High School Diploma	Single	0	Owned	High
4	45	Female	100000	Bachelor's Degree	Married	3	Owned	High

```
df.isna().sum()
# there are no missing values
```

[4] ✓ 0.0s Python

```
Age                 0
Gender              0
Income              0
Education           0
Marital Status      0
Number of Children  0
Home Ownership      0
Credit Score        0
dtype: int64
```

master* ⊗ 0 ⚠ 0 Connect Ln 13, Col 57 Cell 5 of 42

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```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import LabelEncoder
from xgboost import XGBClassifier

from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
from sklearn.metrics import r2_score, mean_squared_error
```

[1] ✓ 16.5s Python

```
df = pd.read_csv('Credit Score Classification Dataset.csv')
```

[2] ✓ 0.0s Python

Generate + Code + Markdown

```
df.head()
```

[3] ✓ 0.0s Python

	Age	Gender	Income	Education	Marital Status	Number of Children	Home Ownership	Credit Score
0	25	Female	50000	Bachelor's Degree	Single	0	Rented	High
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df.isna().sum()
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[4] ✓ 0.0s Python

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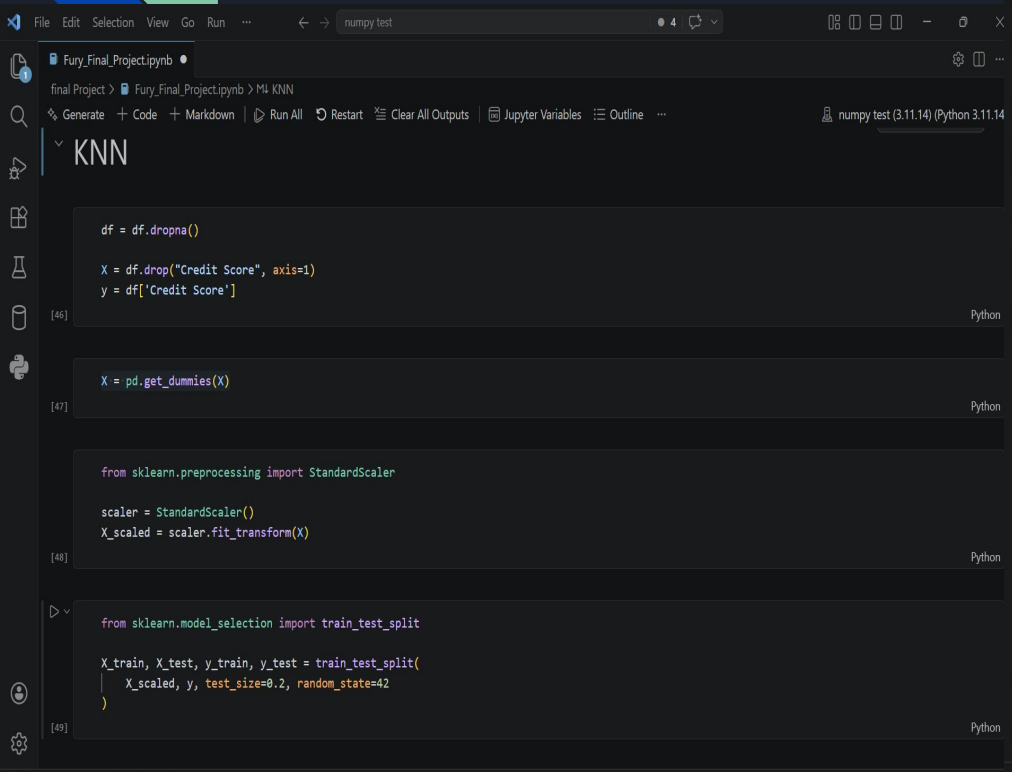
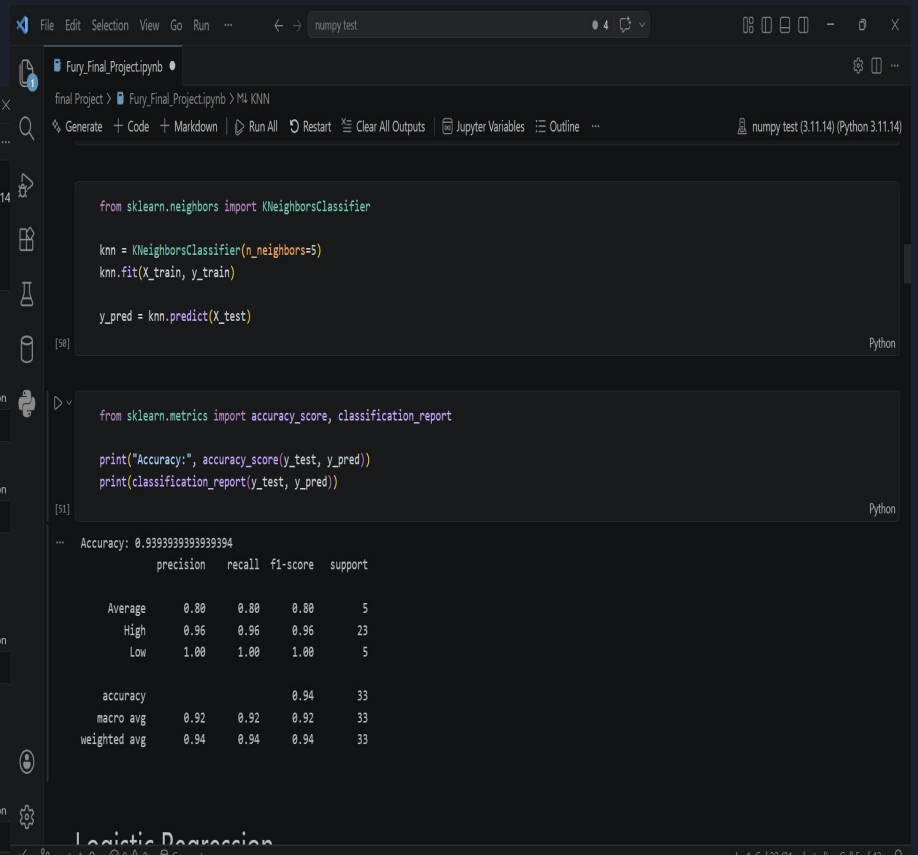
Explanation for all methods

So, for all of the methods I used get_dummies to convert the values into categorical values and train test split to split and train all of the models. Also to help train the models we drop credit score from the x and add it to the y which is what we are trying to predict.

Pictures of the KNN

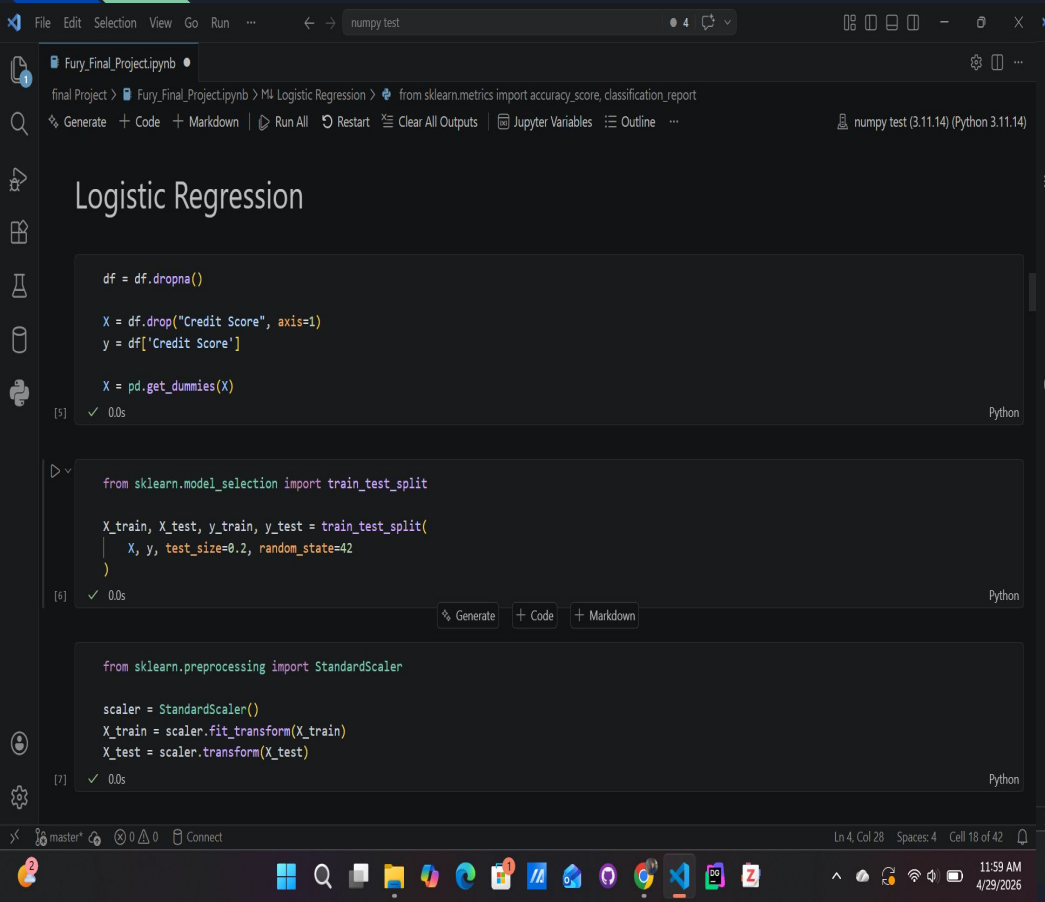


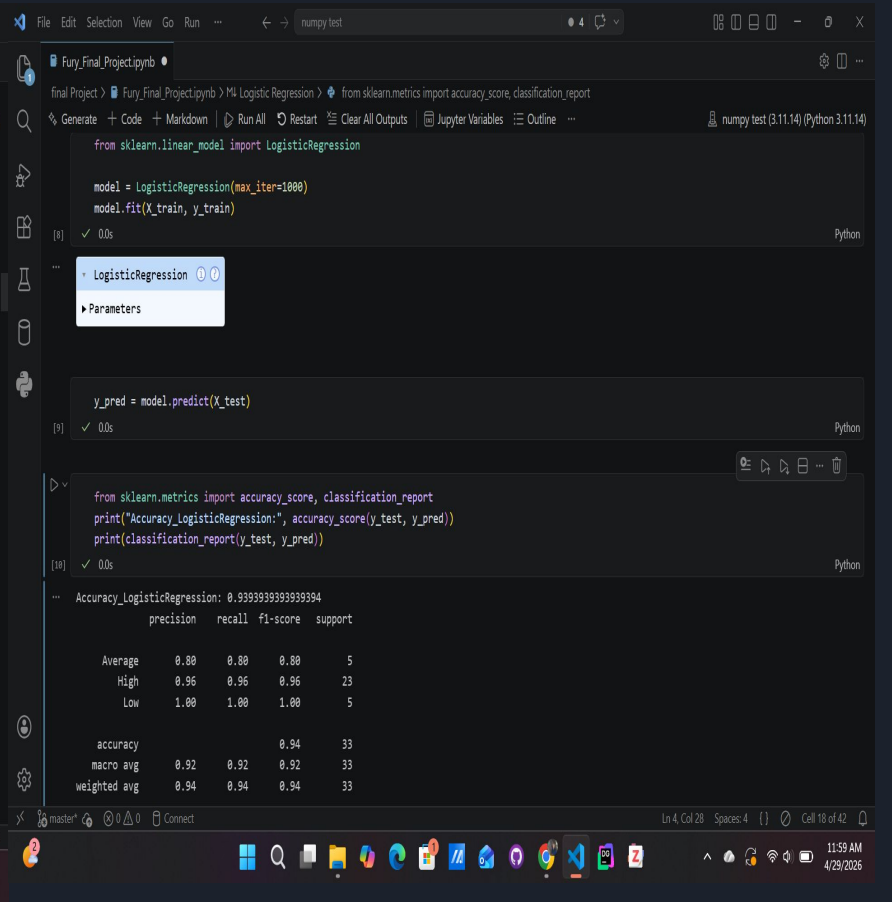


Explanation of KNN

We use standard scaler to scale the data to make sure that every feature is treated fairly to help the model have a better accuracy. Knn uses the knn classifier, and uses n_nighbors.

Pictures of Logistic Regression

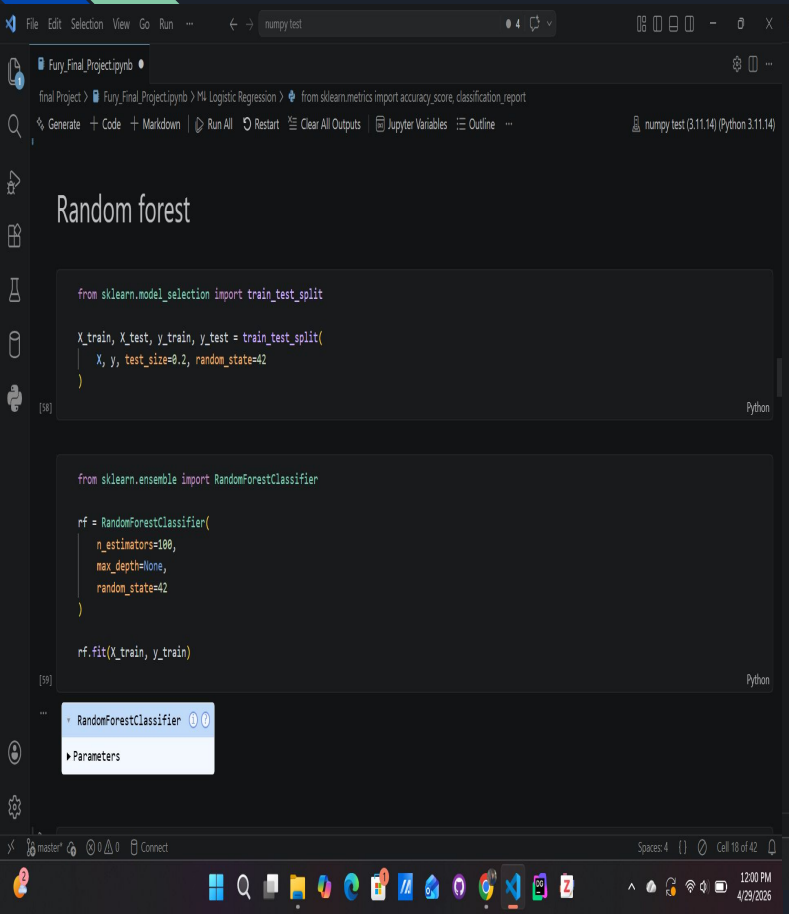


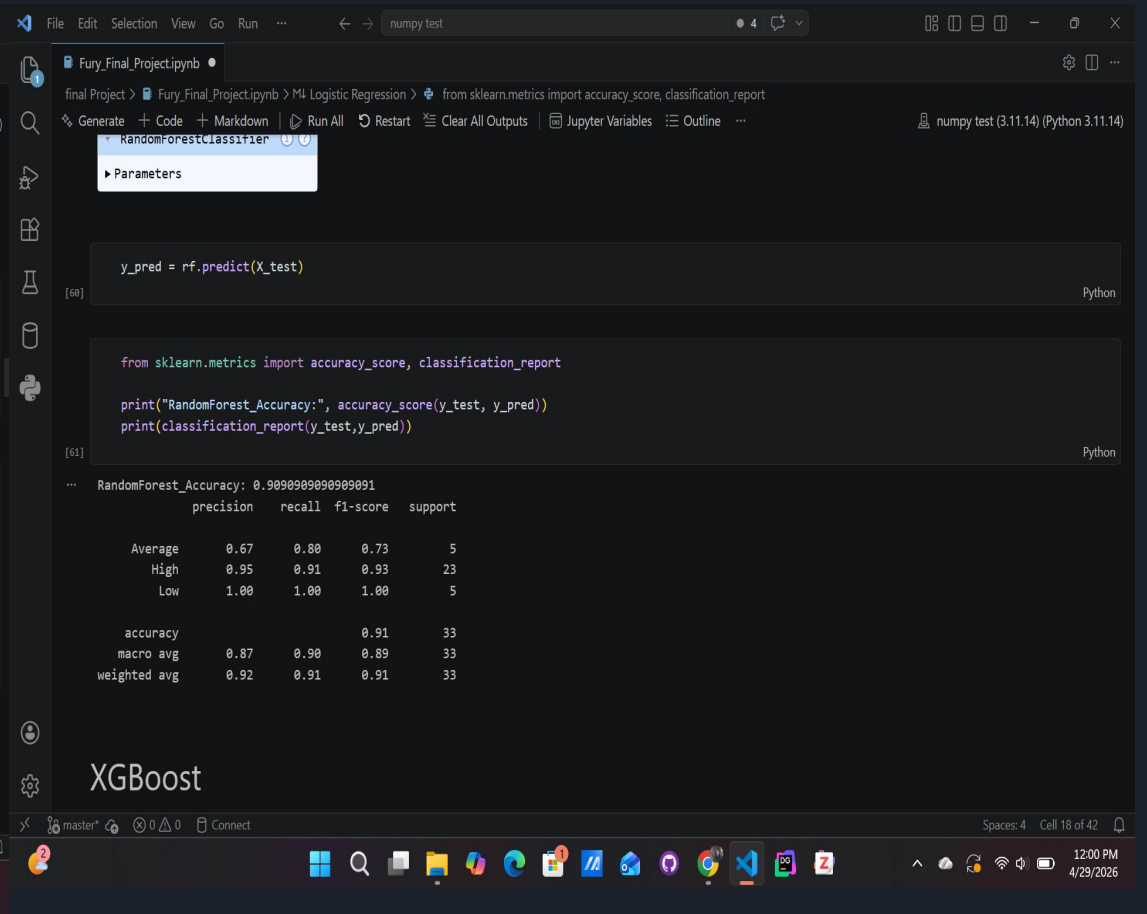


Explanation for Logistic Regression

I used stander scaler again just like KNN to scale the data. For Logistic regression we uses max iterations to predict the score.

Pictures of Random Forest

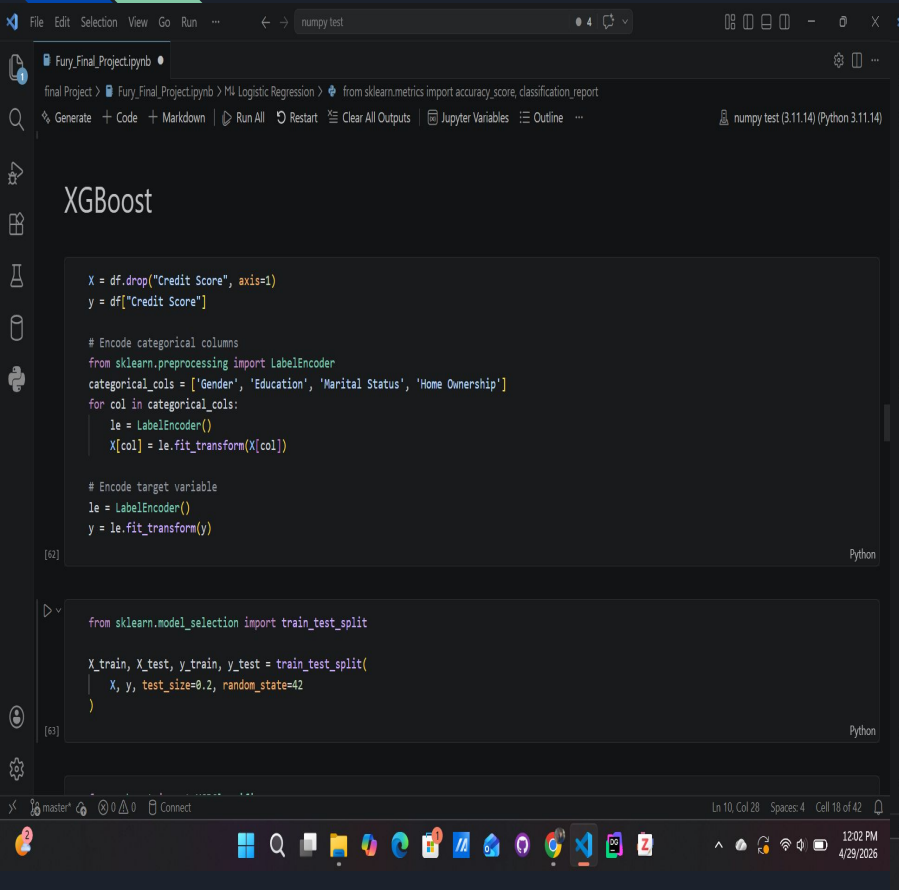


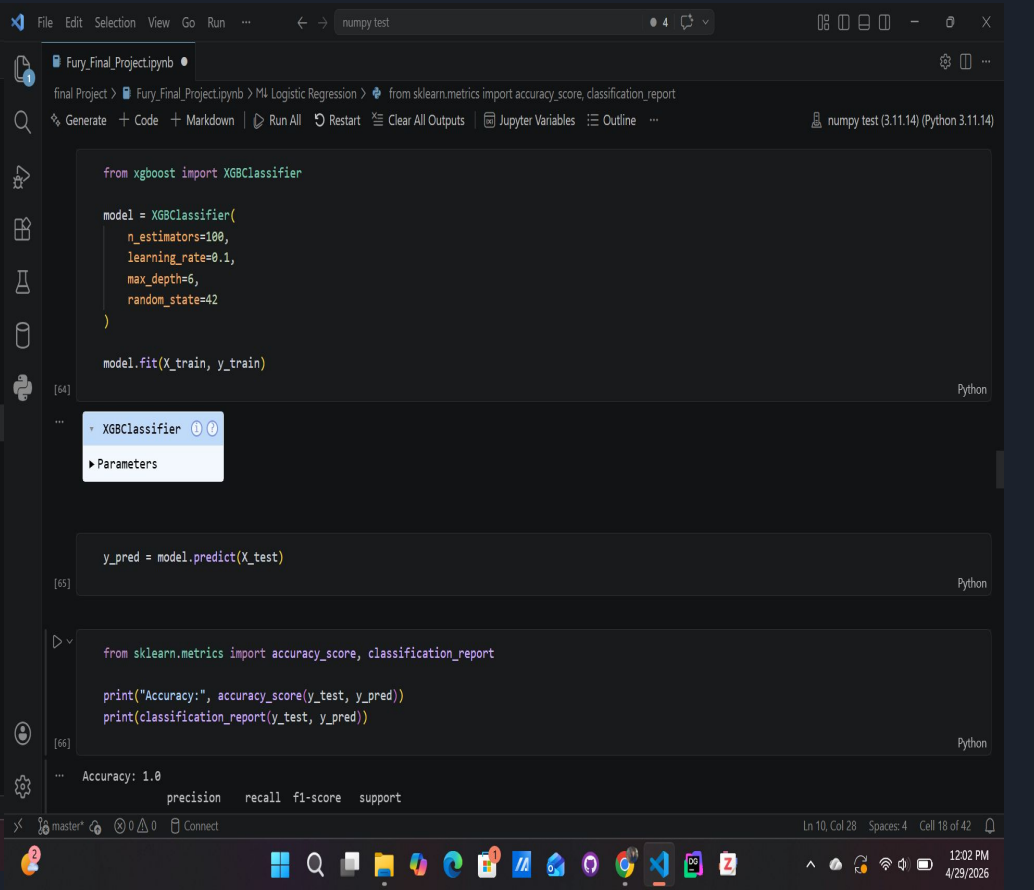


Explanation of Random Forest

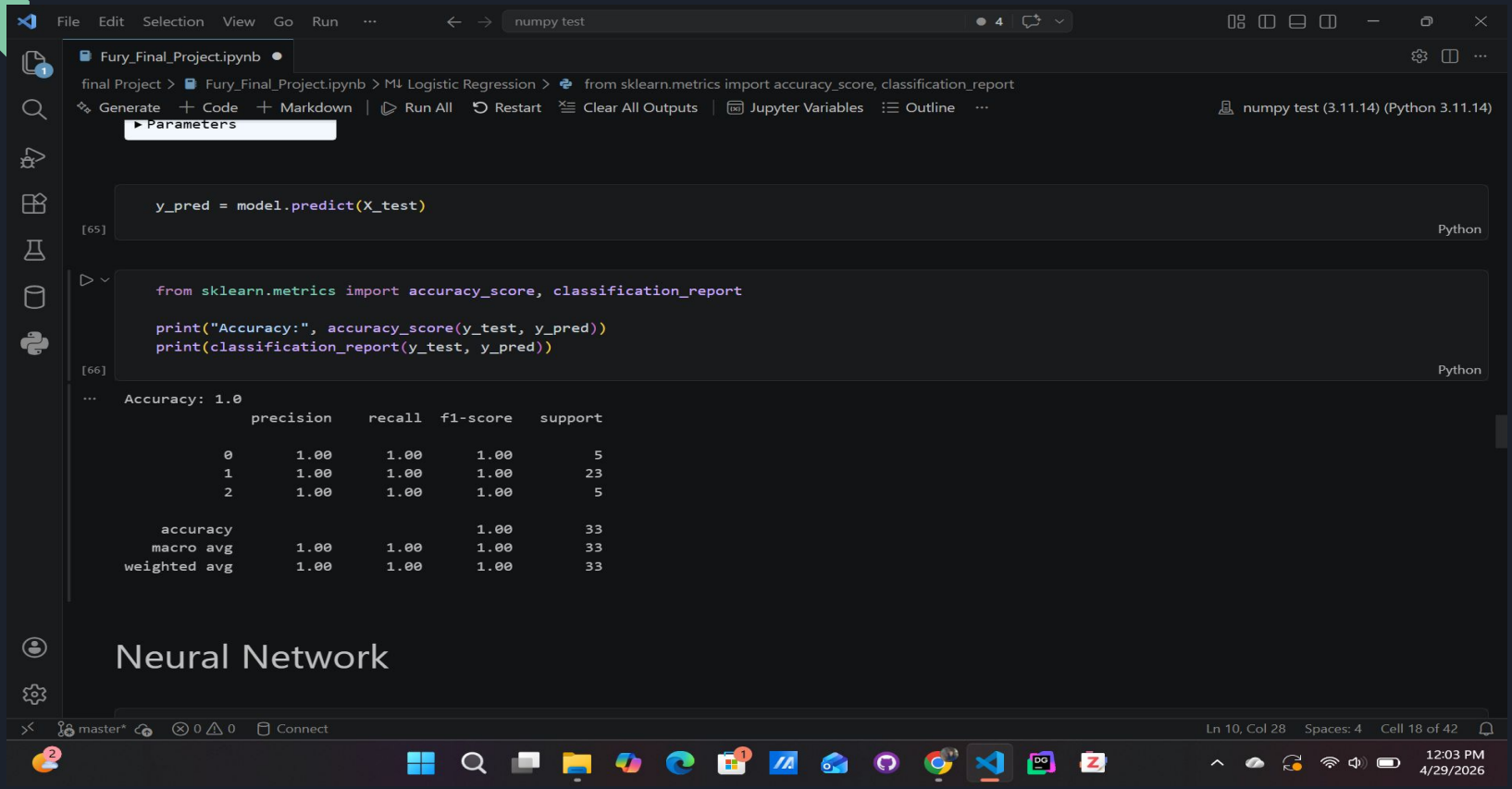
For random forest we use the classifier for random forest witch has the n_estomators, max_depth_ and random state and then we also fit the train data.

XGBoost Pictures





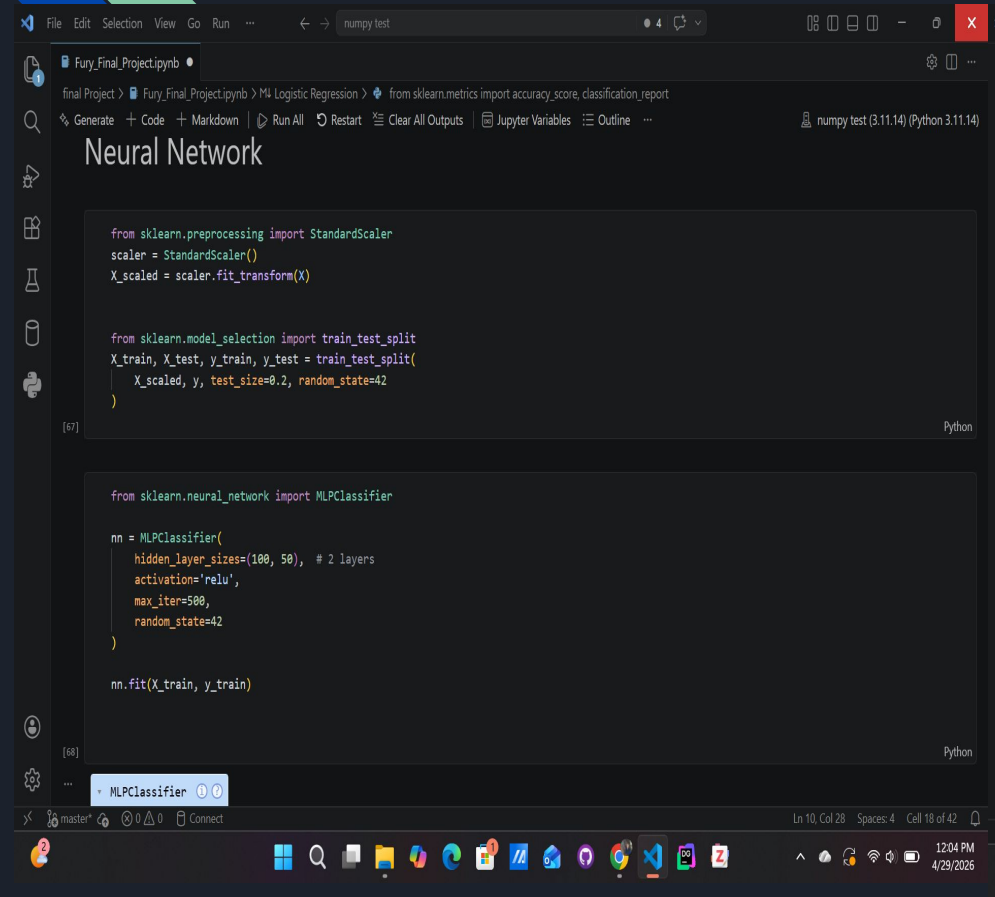
XGBoost Pictures Cont

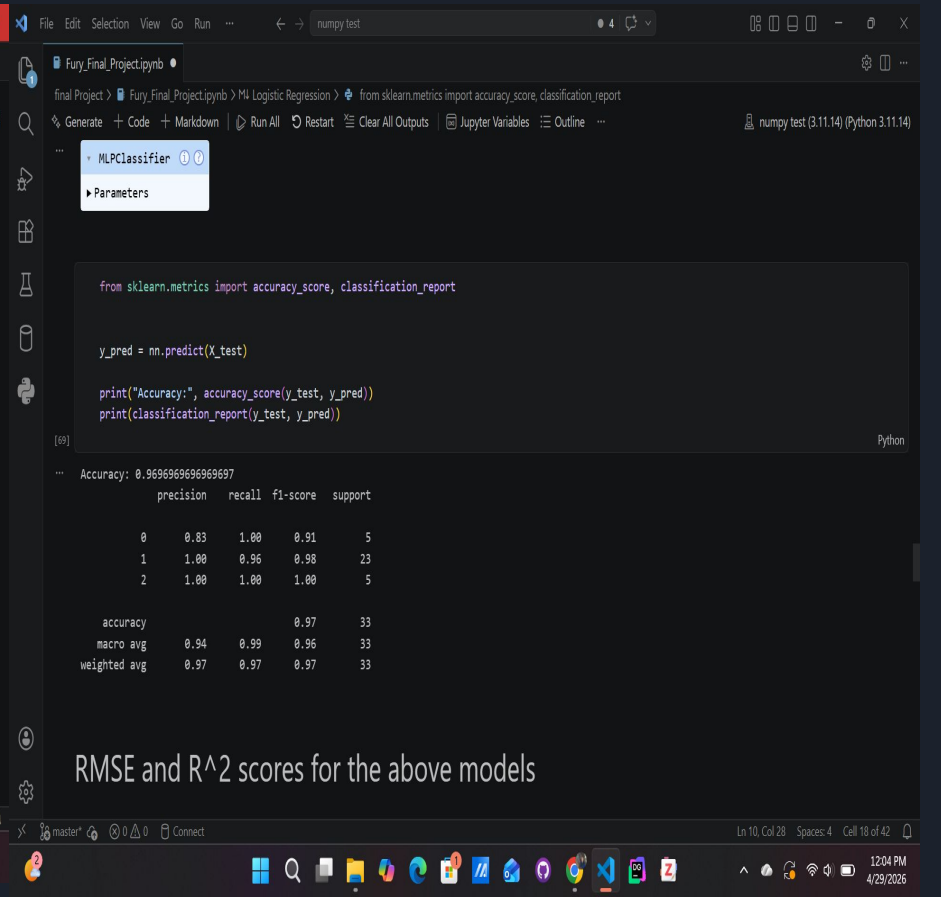


Explanation of XGBoost

For XGBoost we use the Label encoder to put all the column to categorical values, and transforms that data. Then we use the XGBoost classifier which has n_estomators, learning_rate, max_depth_ and ransome state and then fits it.

Neural Network Pictures

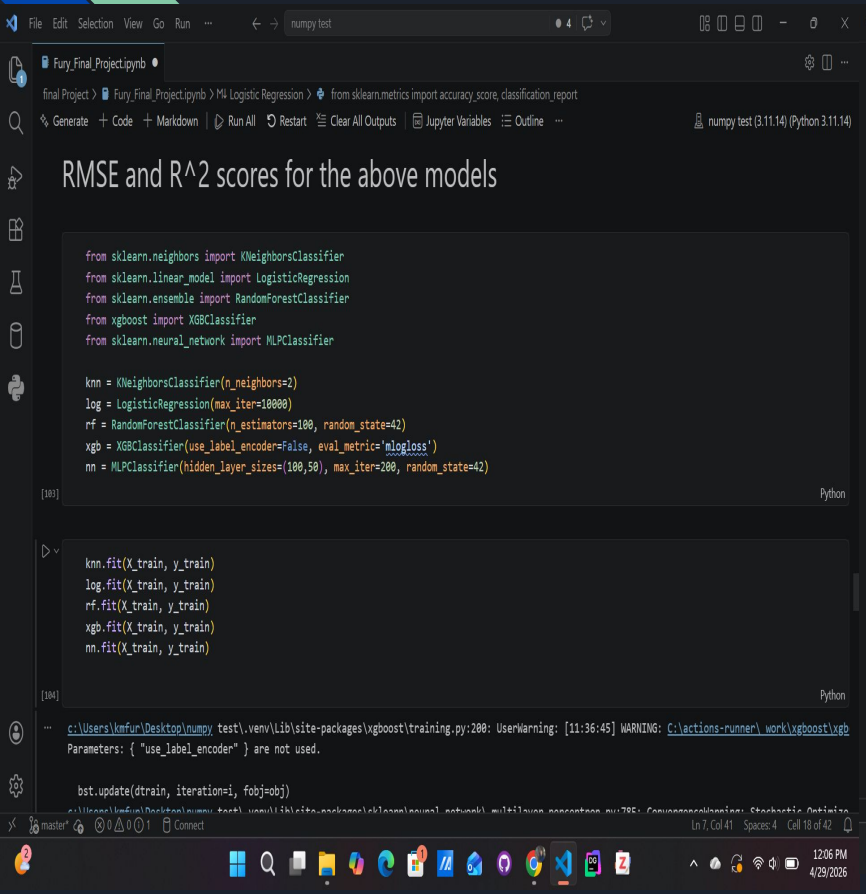


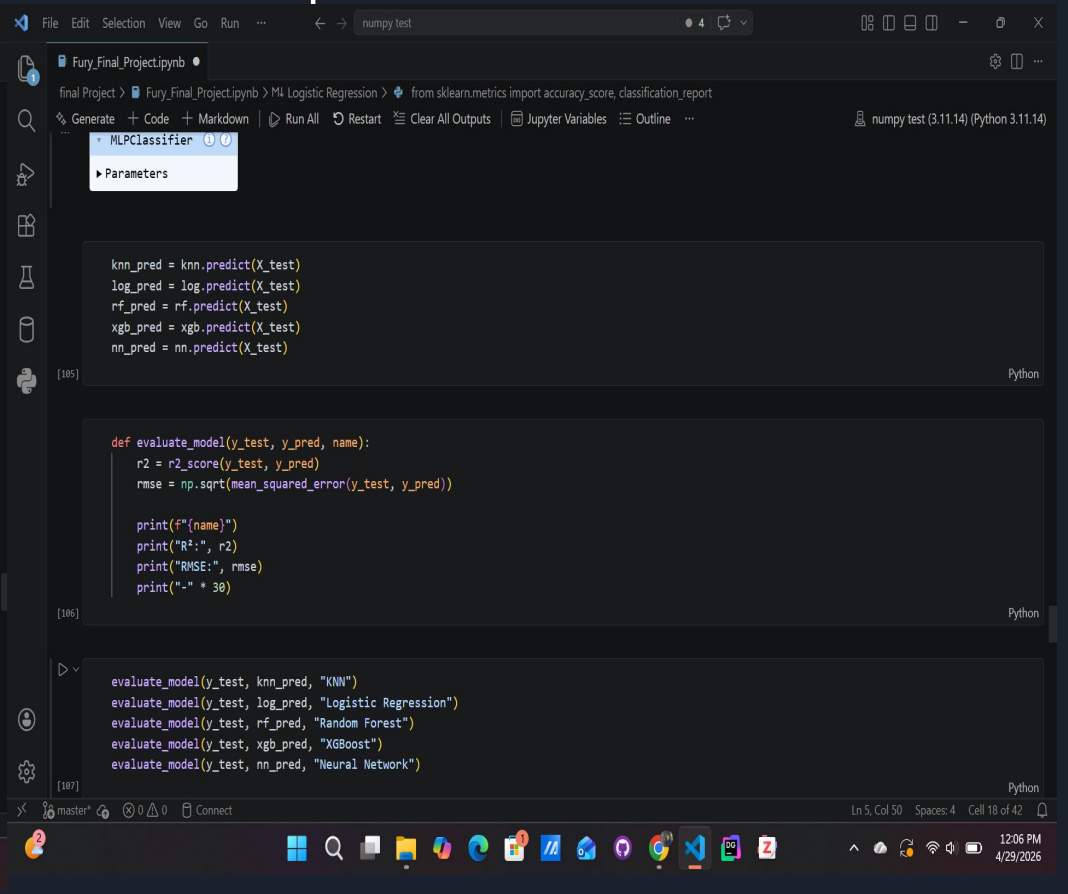


Neural network explanation

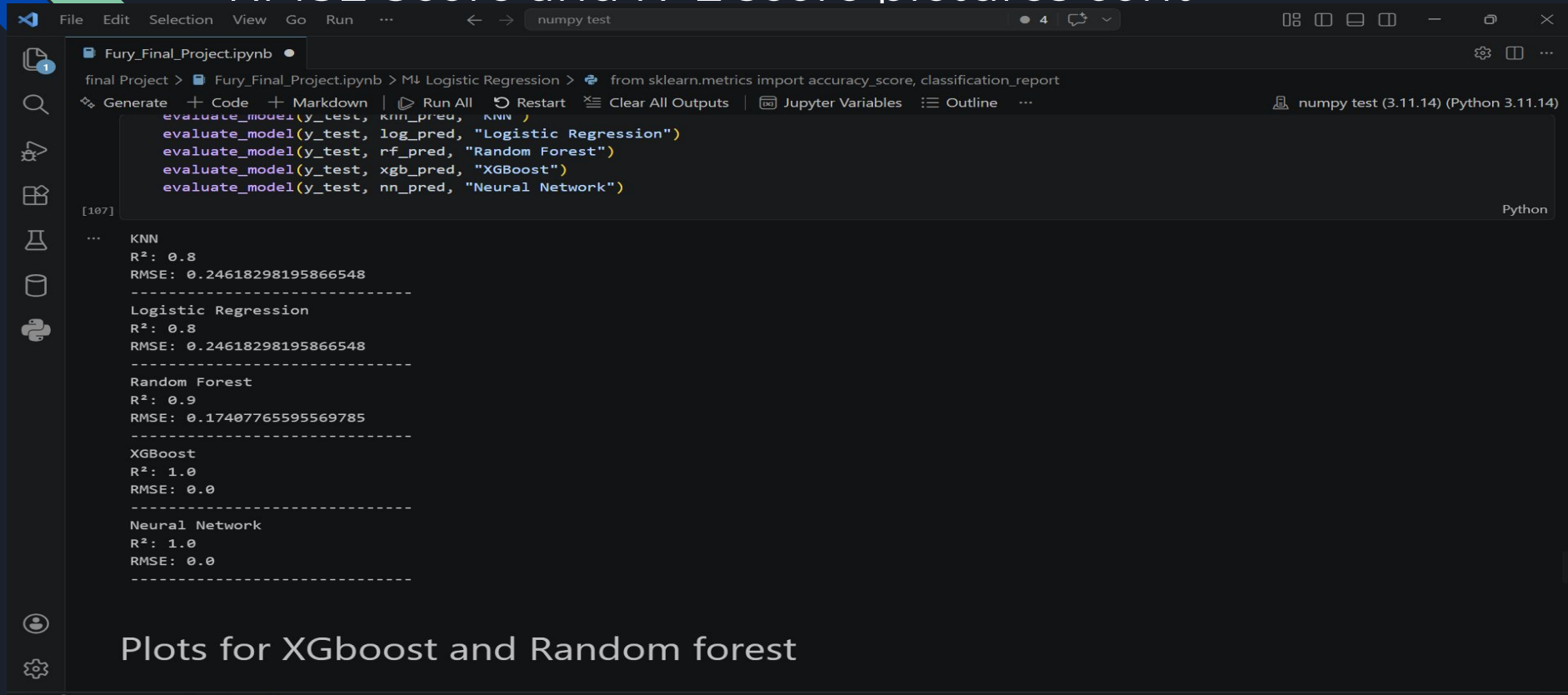
Neural network uses standard scalar just like KNN and Logistic regression. For Neural network we use the MLP classifier which has, hidden_layer_sizes, activation, maxiterations, and random state. Then we fit it.

RMSE Score and R^2 score pictures





RMSE Score and R^2 score pictures cont



Plots for XGboost and Random forest

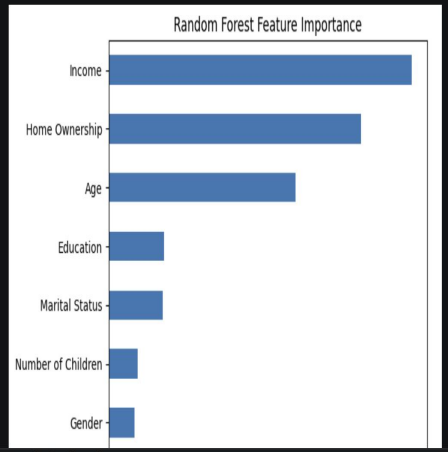
RMSE and R^2 score explanation

We then do the same thing as the accuracy score, but we have to use RMSE and R^2 scores. So we fit the data then predict the data then print out the scores for RMSE and R^2 score.

Plots for XGBoost and Random Forest Pics

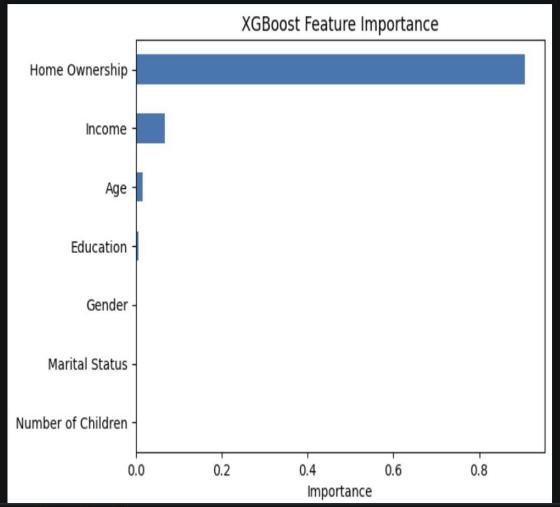
```
import matplotlib.pyplot as plt
rf_importance = pd.Series(rf.feature_importances_, index=X.columns)
rf_importance = rf_importance.sort_values()

plt.figure()
rf_importance.plot(kind='barh')
plt.title("Random Forest Feature Importance")
plt.xlabel("Importance")
plt.show()
```




```
xgb_importance = xgb_importance.sort_values()

plt.figure()
xgb_importance.plot(kind='barh')
plt.title("XGBoost Feature Importance")
plt.xlabel("Importance")
plt.show()
```



Conclusion and issues that was found

This project provided a valuable overview of all semester algorithms. Their accuracy scores were similar, with the neural network performing best. The challenge was determining the optimal RMSE and R-squared values for improved scoring for KNN and Logistic Regression . Other than that it was a good project and had a good time doing it.

Questions, Comments, Concerns?